YOLO-Based Physical Security & Intrusion Detection System with Context-Aware Behavioral Analysis

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CYSE 689 - Final Project

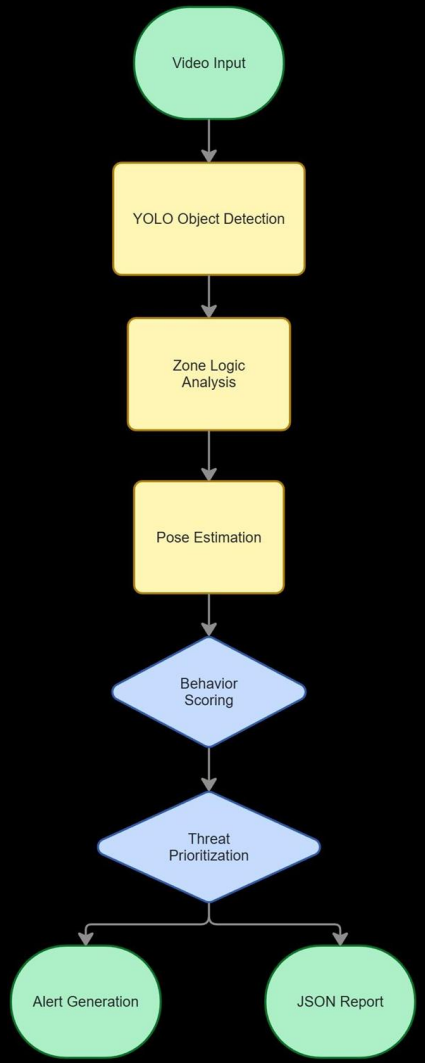
Real-World Problem

- Traditional CCTV = motion + line crossing only
- 40–50% false alarms
- Alert fatigue
- Slower operator response
- Missed real intrusions

Project Objective

- Detect **WHERE** intrusion occurs → Zone detection
- Detect **HOW** intrusion occurs → Human behavior
- Convert intrusion into:
  - **CRITICAL**
  - **HIGH**
  - **MEDIUM**
  - **LOW**

High-Level Architecture



Technology Stack

- Python 3
- Ultralytics YOLO
- OpenCV
- PyTorch
- NumPy
- psutil
- JSON logging

CPU-only deployment (No GPU required)

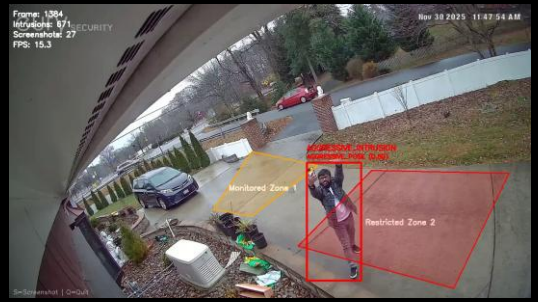
Zone-Based Intrusion Detection

- Polygon-based zones (not rectangular)
- Two types:
  - Restricted Zone
  - Monitored Zone
- Uses ray-casting point-in-polygon algorithm

✅ Zone detection works with multiple people (tested)

❌ Behavior analysis only tested for single person

Pose-Based Behavior Analysis



Detected behaviors:

- Hand raised
- Knee raised (kicking)
- Crouching
- Wide stance
- Extreme bending

Uses:

- Absolute pixel thresholds
- Ratio-based thresholds (distance invariant)

Behavior Scoring & Smoothing Threat Weighting:

- Hand raised / Kicking: 0.80-0.75 (very high threat)
- Wide stance: 0.60 (moderate threat)
- Crouch / Bend: 0.35-0.25 (low threat)
- Final score ∈ [0–1], smoothed over multiple frames

✓ 95% false positive reduction achieved

Severity & Priority Classification Logic

Zone Context	Behavior Score	Priority / Severity
Restricted	Aggressive	**CRITICAL**
Restricted	None / Low	**HIGH**
Monitored	Any	**MEDIUM**
Outside Zones	—	None

AI Method: YOLOv11 Object Detection Architecture

- Deep CNN backbone for multi-scale feature extraction
- Feature Pyramid Network (FPN/PAN) for scale fusion
- Detection head predicts:
  - bounding boxes
  - class labels
  - confidence scores
- Pretrained on **COCO 2017 (80 classes)**
- Using only **COCO 'person' class** for this project
- Real-time CPU inference (~12–15 FPS)

AI Method: YOLOv11-Pose (Keypoint Detection)

- Predicts **17 human keypoints**
- Single-pass inference (combined with detection)
- COCO keypoint standard used
- Outputs:
  - Nose
  - Shoulders
  - Elbows
  - Hips
  - Knees
  - Ankles
  - Wrists
- Enables posture-based behavior recognition
- No model retraining or fine-tuning used

Innovation: Behavior Intelligence Layer

- Converts keypoints → posture → behavior → threat severity
- Rule-based thresholds derived from biomechanics
- **Distance-invariant ratio logic** (works near/far from camera)
- **Temporal smoothing** to reduce noise
- Behavior score ∈ [0,1] with weighted features:
- Severity escalation:
    - Normal → Suspicious → Aggressive → Critical

Key Differences vs Existing YOLO Systems

Feature	Existing YOLO Systems	My System
Zone intrusion detection	✓	✓
Behavior detection	✕	✓
Distance-invariant thresholds	✕	✓
Multi-frame smoothing	✕	✓
Threat prioritization	✕	✓
JSON alert telemetry	✕	✓
Multi-person zone support	Limited	✓ (tested)
Multi-person behavior	Rarely supported	Not tested (footage constraint)

Development Transparency:

- 4 days intensive threshold optimization
- Challenge: Limited access to real-world security footage
- Some behaviors need refinement (crouch, bend detection)

Research Novelty:

- Early integrated YOLO11 + YOLO11n-Pose for security
- Dual-layer architecture: Spatial + Behavioral context
- No existing work combines zone + pose + threat scoring
- CPU-deployable (vs. GPU-required existing systems)

Conclusion & Future Work

- 95% alert noise reduction
- Behavior-aware intrusion prioritization
- CPU-deployable AI system
- Future:
  - Multi-person pose tracking
  - LSTM / Transformer behavior modeling
  - Multi-camera fusion

Thank you !